Technical Verification Companion to the ECH Spin-Torsion Program: ΛCDM+$\Delta N_{\rm eff}$ MCMC Proxy, NaMaster Pipeline Recovery, and a Birefringence Consistency Check with a Spectator-ALP Model

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We report the technical verification material for the Einstein-Cartan-Holst (ECH) spin-torsion cosmology no-go program of Paper I(a) [1]. Three analyses are documented. *(1) Stock-CAMB ΛCDM+$\Delta N_{\rm eff}$ MCMC proxy* (Cobaya v3.6.1, **309,189** frozen samples across two converged dataset combinations; a third Planck-only combination is excluded from all tables and the headline count): this run uses stock CAMB with $\Delta N_{\rm eff}$ as a free parameter and carries *no torsion modifications to the Boltzmann equations*; it is reported as a null-consistency test of an extra radiation-like degree of freedom, not as evidence for or against the ECH spin-torsion framework. Both frozen dataset combinations find $\Delta N_{\rm eff}$ consistent with zero (-0.020 ± 0.169 full-tension; $+0.065\pm0.17$ Planck+BAO+SN) and H_0 consistent with standard ΛCDM (67.68 ± 1.06 full-tension; 67.79 ± 1.09 Planck+BAO+SN, both in $\rm km\,s^{-1}\,Mpc^{-1}$). *(2) NaMaster pseudo-C_ℓ pipeline validation* on synthetic ΛCDM CMB polarization skies with an ACT-like footprint mask ($N_{\rm side}=512$, $\ell_{\max}=1024$, $f_{\rm sky}=0.32$, $10\,\mu\rm K\cdot arcmin$ white noise, 500 Monte Carlo realizations): injecting the spectator-ALP fiducial value $\beta=0.27^\circ$ recovers $\hat\beta=0.238^\circ$ (pipeline-recovery bias $\hat\beta-\beta_{\rm inj}=-0.032^\circ$; the worst-case bias across injections, -0.040° at $\beta_{\rm inj}=0.342^\circ$, is carried forward as the pipeline systematic floor — both are MC pipeline-recovery figures, not sky-measurement systematics, and are not directly comparable to each other's published sky significances). *Scope of the validation*: the test confirms the algebraic pseudo-C_ℓ $E\to B$ deconvolution under MASTER [2] mode coupling, NOT the physical separation of the cosmic-rotation angle β from the instrumental-miscalibration angle α which strictly requires unrotated galactic foregrounds (the synthetic CMB-only skies contain no galactic foregrounds, so the very component that breaks the β–α degeneracy in published Planck/ACT DR6 measurements is absent by construction). The MC recovery is therefore a pipeline-validation figure, not a sky-detection significance claim. The primary sky detection significance is the published Planck/ACT DR6 2.7–2.9σ [3, 4];[a] the pipeline SNR figures refer to recovery of injected MC signals and are *not* competitive sky measurements. *(3) Spectator-ALP consistency check*: a field with $f_a\sim M_{\rm Pl}$ (where $M_{\rm Pl}=(8\pi G)^{-1/2}\approx 2.44\times10^{18}\,$GeV is the *reduced* Planck mass), $m\sim H_0$ is consistent with the published joint WMAP+Planck value $\beta=0.342^\circ\pm0.094^\circ$ (3.6σ) [5]. *Spectator-status caveat*: for $\theta_i\sim1$ the ALP energy density $\rho_a\sim m^2f_a^2\theta_i^2\sim H_0^2M_{\rm Pl}^2$ is of order the critical density today, so the spectator label is only consistent under the explicit restriction $\theta_i\ll1$ (which constitutes fine-tuning of the misalignment initial condition); at $\theta_i\sim1$ the ALP must instead be treated as the dark-energy field itself, which is outside the scope of this companion-paper consistency check. The same birefringence arises in standard GR with an identical ALP; it is not a distinctive ECH prediction, and the spectator-status fine-tuning is required regardless of whether the underlying cosmology is a bounce or ΛCDM. A reproducibility manifest is included in Appendix A.

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[a] Eskilt & Komatsu 2022 disambiguation: the published PRD paper [5] (PRD 106:063503, arXiv:2205.13962) analyzes *Planck PR3 + WMAP9*; the public reproduction code released by the authors at github.com/LilleJohs/Cosmic_Birefringence was subsequently updated to use *Planck PR4 / NPIPE*. Throughout this paper, the labels "PR4/NPIPE" attached to the Eskilt+Komatsu likelihoods refer to the code-repository dataset (which is what the ALP-MCMC re-runs actually use); the abstract $\beta=0.342^\circ\pm0.094^\circ$ (3.6σ) headline is from the published PR3+WMAP9 joint analysis. The repository README is the authoritative source for the dataset attribution in the executed pipeline.

I. INTRODUCTION

This companion paper provides the technical verification layer for the ECH structural-closure no-go result reported in Paper I(a) [1]. The main paper establishes 14 independent structural constraints closing minimal-ECH dark-energy routes and proves a perturbation-transparency theorem for the Holst sector. The present paper documents the three numerical analyses that support and contextualize those results.

Scope of this paper.—Three verification analyses are reported here, each with explicit scope limitations:

1. *ΛCDM+$\Delta N_{\rm eff}$ MCMC proxy* (Secs. II and III): stock CAMB Boltzmann solver with $\Delta N_{\rm eff}$ as a free parameter. **Not a spin-torsion theory module** (see the §III scope statement); this run tests whether the data prefer an extra radiation-like degree of freedom in standard cosmology.

2. *NaMaster CMB E-B analysis* (Sec. IV): a bias-injection Monte Carlo validation of the pseudo-C_ℓ deconvolution pipeline on synthetic ΛCDM polarization skies. **Not a competitive sky detection.** The high pipeline template-fit SNR figures (e.g., 20.32) refer to recovery of injected MC signals, not to the significance of the CMB sky measurement, which remains the published 2.7–2.9σ from Planck/ACT.

3. *Spectator-ALP consistency check* (Sec. VI): a standard GR+ALP computation showing that the observed signal is consistent with an ALP having natural parameters (taken at scan-prior midpoint values; the $\sim 25\times$ misalignment tuning required to reconcile the headline result with the spectator-consistent corner is disclosed in Sec. VI and fn. 5). **Not a distinctive ECH prediction.** The same $\beta \approx 0.27°$ arises in any GR+ALP setup with the same parameters; no ECH-specific derivation connects the Holst action to the photon-torsion coupling required.

What is NOT in this paper.—The 13 logically-independent structural barriers, the perturbation-transparency theorem, the 14-barrier table, and the surviving matter-bounce-specific test predictions ($f_{\rm NL} = -35/8$, valid strictly for the minimal scalar-only $w = 0$ matter-dominated contraction class, not the broader bouncing-cosmology landscape which encompasses ekpyrotic, Cuscuton, string-gas, and quintom variants with different $f_{\rm NL}$ signatures, and ALP birefringence) are in Paper I(a) [1]. The SPHEREx multi-tracer Fisher forecast is in Paper II [6]. The multi-survey anomaly catalog is in Paper III [7]. The galaxy chirality catalog is in Paper IV [8].

Cross-paper citations.—When this companion reports MCMC values (H_0, σ_8, etc.) that are referenced in the main paper, those values come from Secs. III and V here.

II. COSMOLOGICAL TENSIONS: H_0 AND σ_8

The bounce scenario motivates extending ΛCDM by $\Delta N_{\rm eff}$ (particle production at the bounce) as a phenomenological proxy parameter; $(\omega/H)_0$ (angular momentum transfer) and Ω_k are fixed to zero in the actual sampled MCMC configuration of this paper (per the explicit parameter-scope clarification in Sec. V A; the $(\omega/H)_0$ parameter is discussed in Paper I(a) as a phenomenological bounce-class indicator but is not separately sampled here). The full-tension dataset combination includes the SH0ES [9] M_B-anchor likelihood (`H0.riess2020Mb`, entering through the supernova absolute magnitude rather than as a direct H_0 prior) in the Cobaya likelihood configuration, but the high-precision Planck NPIPE (PR4) CamSpec high-ℓ TTTEEE + Planck 2018 low-ℓ TT/EE + Planck 2018 lensing likelihoods carry sufficient inverse-variance weight that the posterior H_0 in the proxy run is pulled to 67.68 ± 1.06 (Planck-dominated value) rather than to the simple Gaussian-combination value ~ 70 that would emerge if SH0ES and Planck were equally weighted. We do not therefore claim that the SH0ES tension is resolved or even moved by adding $\Delta N_{\rm eff}$ in stock CAMB; in this configuration the $\Delta N_{\rm eff}$ posterior is consistent with zero and the H_0 posterior is dominated by Planck. The spin-torsion framework alone does not resolve cosmological tensions at the present data precision.

III. STOCK-CAMB ΛCDM+$\Delta N_{\rm eff}$ MCMC: GENERIC RADIATION-PROXY TEST (NOT A SPIN-TORSION THEORY MODULE)

Scope statement.—"MCMC verification" refers throughout to a stock CAMB run of ΛCDM extended by $\Delta N_{\rm eff}$ as a free parameter. *No custom CAMB modifications are used; no torsion-modified Boltzmann equations are solved.* The MCMC therefore tests whether the data prefer an extra radiation-like degree of freedom, treated as a generic phenomenological proxy for the spin-torsion sector's possible effective radiation contribution. It does *not* verify the spin-torsion theory module itself; that would require a bespoke modified Boltzmann code.

The proxy run (Cobaya v3.6.1 with Planck NPIPE (PR4) CamSpec high-ℓ TTTEEE + Planck

2018 low-ℓ TT/EE + Planck 2018 lensing likelihoods, i.e. `planck_NPIPE_highl_CamSpec.TTTEEE` + `planck_2018_lowl.TT` + `planck_2018_lowl.EE` + `planck_2018_lensing.clik` in the Cobaya YAML) has produced two frozen dataset combinations with publication-quality convergence (limitation: the PR4/NPIPE high-ℓ + 2018-release low-ℓ/lensing mixture is the standard Cobaya pairing, but we have not run a release-pairing swap test with PR4-consistent low-ℓ and lensing counterparts; any pairing-induced bias on the headline $\Delta N_{\rm eff}/H_0/S_8$ at the quoted precision is therefore unquantified here), plus a third Planck-only run currently at sub-convergence sample count and not aggregated into any frozen-posterior summary statistic in this paper. Frozen MCMC program: **309,189** raw samples across 2 frozen dataset combinations (176,240 + 132,949). We report this as a ΛCDM+$\Delta N_{\rm eff}$ null-consistency test: the data are consistent with $\Delta N_{\rm eff} = 0$ in stock CAMB. Sampling configuration (per the on-disk Cobaya YAMLs): $\Delta N_{\rm eff}$ enters as `nnu` with a flat prior $N_{\rm eff} \in [2.046, 5.046]$ (i.e. $\Delta N_{\rm eff} \in [-1, +2]$); the neutrino mass is held at the CAMB default $\Sigma m_\nu = 0.06\,$eV with one massive eigenstate (`num_massive_neutrinos`=1), and $Y_{\rm He}$ follows the CAMB BBN-consistent default (no explicit override in any of the four YAMLs). The prior allows negative $\Delta N_{\rm eff}$ (down to -1), so the results are two-sided posterior means, not physical "extra-species" limits. Under the physically motivated restriction $\Delta N_{\rm eff} \geq 0$, the one-sided 95% upper limit from the full-tension chain is $\Delta N_{\rm eff} < 0.31$ (read from the posterior CDF at the 95$^{\rm th}$ percentile of the $\Delta N_{\rm eff} > 0$ half; the planck+BAO+SN chain gives $\Delta N_{\rm eff} < 0.39$). These bounds are consistent with the two-sided means quoted throughout; both conventions are reported because a published-extra-species interpretation of the run requires the one-sided limit.[1]

a. Scope of the $\Delta N_{\rm eff}$ proxy: bounce-class discrimination, not a direct test of the spin-torsion sector. We emphasize that the stock-CAMB ΛCDM+$\Delta N_{\rm eff}$ proxy run does *not* test the ECH spin-torsion sector directly. The Hehl-Datta-Mercuri parity-even four-fermion contact interaction that survives torsion elimination [10, 11] is dimension-6 and $M_{\rm Pl}^{-2}$-suppressed[2]: its leading Boltzmann effect is a scattering-amplitude shift, not a relativistic species, and it does *not* produce a $\Delta N_{\rm eff}$ at recombination. The *minimal* matter-bounce class [12] (here "minimal" = the single-scalar $w = 0$ matter-dominated contraction phase, distinguished from the broader matter-bounce family including ekpyrotic, cyclic, Cuscuton, string-gas, and quintom-bounce variants) predicts $\Delta N_{\rm eff} \approx 0$ by construction (no light bounce-internal species are thermalized at recombination); the proxy run *confirming* $\Delta N_{\rm eff} = -0.020 \pm 0.169$ (full-tension) and $+0.065 \pm 0.17$ (Planck+BAO+SN) is therefore consistent with the minimal matter-bounce prediction, but the absence of a $\Delta N_{\rm eff}$ detection is not a discriminator between minimal-ECH and standard ΛCDM at the present data precision. We frame the proxy as a bounce-class *compatibility check* (minimal matter-bounce-class scenarios without prolonged post-bounce inflation predict $\Delta N_{\rm eff} \approx 0$), not as a posterior-preference test against a competing model.

Physics interpretation (Table II).—The converged $w_0 w_a$ posterior *disfavors* (in the marginal-tail sense; see fn. a) the ΛCDM point $(w_0, w_a) = (-1, 0)$ at the joint level: w_0 departs by $+4.3\sigma$ and w_a departs by -3.6σ, with $w_0 + w_a = -1.48 \pm 0.15$ requiring phantom crossing in the redshift range probed by DESI DR2 BAO + DES-Y5 + Pantheon+: at the posterior mean the CPL trajectory $w(a) = w_0 + (1-a)\,w_a$ crosses $w = -1$ at $1 - a_\times = (-1 - w_0)/w_a = 0.282$, i.e. $z_\times \approx 0.39$, interior to both the BAO and SN redshift coverage (phantom at $z > z_\times$, quintessence-like today). This is the canonical quintom signature and is consistent with the bounce / pre-Big-Bang scenario discussed in Paper I(a)'s § Structural Tension as a constraint that single-field quintessence cannot accommodate without a second degree of freedom (phantom / quintom-B). In the converged chain there are zero free-$w_0 w_a$ samples at the ΛCDM

[1] Sample-count stratification (reconciliation): **309,189** is the sum of the two frozen combinations ($176{,}240 + 132{,}949$ raw accepted samples). After removing the first 30% of each chain as burn-in, $176{,}240 \times 0.7 + 132{,}949 \times 0.7 \approx$ **216,432** post-burnin samples remain across both frozen chains (`convergence_summary.json`). For the full-tension subset specifically, $176{,}240 \times 0.7 \approx$ **123,368** post-burnin (the **119,617** figure in Fig. 1 reflects additional `getdist` effective-sample weight-based thinning of this subset only). The post-burnin count of the full-tension subset alone is 123,129 (within ±1% of the 123,368 exact computation, with the small offset reflecting the chain-end-truncation of partial samples at the burn-in cut); the correct both-chains post-burnin total is 216,432. *Burn-in reconciliation note:* The frozen `planck_bao_sn_20260312_1954 convergence_report.txt` reports "Burn-in: 20%" and post-burnin samples = 106,361, reflecting the per-chain GetDist readout that averages the parallel chains (`burn_in: 0.1` in their `cobaya_config.yaml`) with the original chain (`burn_in: 0.3`). The 30% figure used throughout this paper is conservative, matches the original-chain configuration documented in `COUNT_EXPLANATION.md`, and gives 93,064 post-burnin for this subset. The 106,361 figure at 20% is the GetDist-reported value; the paper's 216,432 combined total uses the conservative 30% cut uniformly for both frozen chains. The third (Planck-only) dataset combination (114,992 raw samples; $\hat{R} - 1 \sim 0.05$) is still accumulating samples, is *not* reported in Table I (which contains only the two frozen combinations), and is *not* aggregated into the 309,189-sample headline anywhere in this paper.

[2] The dimension-6 four-fermion contact operator is interpreted here in the low-energy effective field theory (EFT) below the strong-coupling / torsion-resolution scale $\Lambda_{\rm strong} \sim M_{\rm Pl}/\sqrt{\gamma_{\rm BI}}$ set by the inverse Barbero–Immirzi parameter $\gamma_{\rm BI}$ (cf. [11], where the contact interaction is derived; the strong-coupling scale itself is a heuristic EFT-validity estimate); above this scale the contact-operator description breaks down and the full Holst-sector dynamics with propagating torsion must be retained. All recombination-era and late-time observables addressed in this paper sit at energies $E \ll \Lambda_{\rm strong}$, where the EFT treatment is controlled.

TABLE I. ΛCDM+$\Delta N_{\rm eff}$ proxy MCMC results from frozen Cobaya v3.6.1 chains (CAMB v1.6.5, stock; *no torsion modifications*). All values are posterior means $\pm 1\sigma$. Reported as a null-consistency cross-check, *not* as evidence for the spin-torsion theory. $S_8 \equiv \sigma_8 (\Omega_m/0.3)^{1/2}$, computed as a Cobaya derived parameter with exactly this definition. The full-tension column includes the DES-Y3 S_8 Gaussian prior 0.776 ± 0.017 (`cobaya_full_tension.yaml`). The Planck+BAO+SN S_8 marginal is 0.827 ± 0.010 from a direct GetDist pass over the frozen chains (132,949 samples). With the chain-recomputed marginal, the full-tension posterior $S_8 = 0.814 \pm 0.008$ is consistent with the naive two-Gaussian combination of the Planck+BAO+SN marginal and the DES-Y3 prior ($0.827 \pm 0.010 \otimes 0.776 \pm 0.017 = 0.814 \pm 0.009$; agreement at the 0.01σ level). The Planck+BAO+SN marginal sits in 2.6σ two-Gaussian tension with DES-Y3 (posterior-overlap integral $\int \min(p_1, p_2)\, dS_8 = 0.05$); the full-tension posterior sits at 2.0σ from DES-Y3 (overlap 0.12); overlay artifact `reproducibility/cosmology/ c13_s8_desy3_overlay.json`.

Parameter	Full-tension	Planck+BAO+SN
H_0 [km s^{-1} Mpc^{-1}]	67.68 ± 1.06	67.79 ± 1.09
$\Delta N_{\rm eff}$	-0.020 ± 0.169	$+0.065 \pm 0.17$
σ_8	0.803 ± 0.008	0.812 ± 0.009
S_8	0.814 ± 0.008	0.827 ± 0.010
Ω_m	0.308 ± 0.005	0.312 ± 0.006
τ	0.054 ± 0.007	0.056 ± 0.007
n_s	0.965 ± 0.006	0.967 ± 0.006
Chains	6	6
Total samples	176,240	132,949
Worst $\hat{R} - 1$[a]	0.001	0.003
Min ESS	4,744	4,692

[a] Worst row is n_s in the full-tension combination at $\hat{R} - 1 = 9.74 \times 10^{-4}$; all sampled parameters across both frozen combinations satisfy $\hat{R} - 1 < 3 \times 10^{-3}$. The full-tension chain samples 17 parameters: 7 cosmological + 9 Planck likelihood nuisance ($A_{\rm planck}$, amp$_{143}$, amp$_{217}$, amp$_{143\times217}$, n_{143}, n_{217}, $n_{143\times217}$, calTE, calEE) + 1 supernova absolute magnitude (M_b, entering via the Pantheon+ SN and SH0ES `H0.riess2020Mb` likelihoods rather than the Planck likelihood stack). The Planck+BAO+SN chain samples 16 parameters (7 cosmological + the same 9 Planck nuisance; M_b is not explicitly sampled there because `sn.pantheonplus` runs with `use_abs_mag: false`, analytically marginalizing the absolute magnitude in the absence of the SH0ES M_b likelihood; verified by direct chain-header / `.updated.yaml` blocking-list inspection). Full per-parameter convergence tables ($\hat{R}$, ESS, drift, per dataset) are archived at `reproducibility/cosmology/convergence_latest.csv` and mirrored in the HuggingFace chain-diagnostics dataset (Appendix A).

point; the chain is centered well into quintom-B territory at $w_0 + w_a \approx -1.48$. To be explicit about what is and is not claimed: this companion reports posterior distance from ΛCDM in the marginalized-tail sense only; rigorous ΛCDM-vs-quintom-B *model preference* ($\ln B$, ΔAIC/BIC) requires the deferred nested-sampling analysis and is *not* claimed here. At the decorrelation pivot ($z_p = 0.27$, a redshift — numerically coincident with, and unrelated to, the §IV injection angle $\beta = 0.27°$; fn. b), the effective equation of state is $w_{\rm pivot} = -0.952 \pm 0.019$, i.e. $+2.5\sigma$ above -1: the departure from ΛCDM persists, at reduced significance, even in the best-determined direction of the (w_0, w_a) posterior, with the largest single-axis departures carried by w_0 ($+4.3\sigma$) and the time-varying w_a term (-3.6σ).

Caveats.—(a) The robust Bayesian evidence / Bayes factor $\ln B$ against ΛCDM is NOT reported here; standard posterior MCMC samples do not give a robust $\ln B$, and a Savage-Dickey density ratio at the ΛCDM point is *not* viable on this chain because the ΛCDM point $(w, w_a) = (-1, 0)$ lies at $> 4\sigma$ in the joint marginal tails (Table II: w_0 departs by $+4.3\sigma$ and w_a by -3.6σ) and is therefore unsampled by the Metropolis-Hastings chain; any KDE-based Savage-Dickey ratio at an unsampled point yields arbitrary kernel-dependent noise rather than a controlled posterior density (with zero free-$w_0 w_a$ samples at the ΛCDM point, a KDE estimator on the converged 2D (w, w_a) marginal fails catastrophically). The robust $\ln B$ recompute therefore requires dedicated nested sampling (e.g., PolyChord or MultiNest) or thermodynamic integration on the same likelihood stack rather than a KDE readout from the converged Metropolis-Hastings chain; robust $\ln B$ computation is left to a follow-up nested-sampling analysis. (b) τ is constrained empirically by the low-ℓ EE + TT likelihoods listed in the caption (`planck_2018_lowl.EE` + `planck_2018_lowl.TT`) and is sampled as a free parameter rather than fixed by a Gaussian prior at the Planck best-fit value (the chain has τ free with the standard low-ℓ data constraint, not a Gaussian prior at the Planck best-fit). (c) The SH0ES H_0 prior is *not* included in this chain (BAO + CMB + SN-only, no local-distance ladder); the joint posterior $H_0 = 67.185 \pm 0.455$ km/s/Mpc (the iter2 $w_0 w_a$ chain of Table II, to which this caveat list refers — not the ΛCDM+$\Delta N_{\rm eff}$ chains of Table I, whose Planck+BAO+SN value is 67.79 ± 1.09) is therefore the no-SH0ES result. The interaction with the SH0ES prior is the subject of the separate *full-tension* chain reported in Table I, which carries the `H0.riess2020Mb` likelihood active in its YAML (verified by direct `.input.yaml` inspection and by chain sample-mean readout: $M_B = -19.263 \pm 0.049$ mag, agreeing with the Riess+2020 SH0ES value $M_B = -19.253 \pm 0.027$ mag at 0.2σ). The full-tension chain returns $H_0 = 67.68 \pm 1.06$ km/s/Mpc with $\Delta N_{\rm eff} = -0.020 \pm 0.169$, exhibiting the canonical 3.6σ Hubble tension with Riess $H_0 = 73.04 \pm 1.04$ km/s/Mpc that the ΛCDM+$\Delta N_{\rm eff}$ extension is unable to resolve: the `H0.riess2020Mb` likelihood constrains the SN Ia absolute magnitude M_B (which is correctly pulled toward the Riess value), but H_0 is determined predominantly by the BAO+CMB acoustic scale through Pantheon+, and the $\Delta N_{\rm eff}$ extension lacks the degrees of freedom to shift H_0 enough to accommodate both data sets. Direct YAML inspection confirms the SH0ES likelihood is active and the M$_B$ posterior is correctly pulled; the 3.6σ tension persists because the model cannot shift H_0 enough to satisfy both data sets simultaneously — the canonical Hubble-tension result, not a YAML omis-

TABLE II. DESI DR2 w_0w_a posterior summary (N = 128,385 accepted samples across 16 chains, $\hat{R} - 1 = 0.00820$; 8 cosmological + 9 nuisance parameters). Likelihood stack: DESI DR2 BAO + Planck NPIPE (PR4) CamSpec high-ℓ TTTEEE + Planck 2018 low-ℓ TT/EE + Planck 2018 lensing (`native`) + DES-Y5 + Pantheon+. Full parameter list and chain diagnostics are available in the companion data repository. Derived-parameter widths are GetDist readouts of the full chain covariance: the chain's σ_8–Ω_m correlation is $\rho = -0.27$ (the standard CMB acoustic-scale/lensing degeneracy), which is why $\sigma_{S_8} = 0.0089$ is smaller than the uncorrelated propagation through $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$ (≈ 0.010).

Parameter	Mean $\pm\sigma$	vs ΛCDM
Dark-energy extension		
w_0	-0.8122 ± 0.0436	(marg.-tail, $+4.3\sigma$)[a]
w_a	-0.6666 ± 0.1864	-3.6σ from 0 (marg.-tail; fn. a)
$w_0 + w_a$	-1.4788 ± 0.1485	phantom-crossing required
$w_{\rm pivot}$	-0.952 ± 0.019	$+2.5\sigma$ from -1[b]
Standard cosmology		
H_0 [km s^{-1} Mpc^{-1}]	67.185 ± 0.455	—
Ω_m	0.3142 ± 0.0045	—
σ_8	0.8057 ± 0.0083	—
S_8	0.8245 ± 0.0089	—
n_s	0.9655 ± 0.0036	—
$10^9 A_s$	2.087 ± 0.030	—
τ	0.0537 ± 0.0072	—
$\omega_b h^2$	0.02224 ± 0.000125	—
$\omega_c h^2$	0.11892 ± 0.000819	—
$100\theta_{\rm MC}$	1.04087 ± 0.000239	—
Age [Gyr]	13.763 ± 0.019	—
Goodness-of-fit decomposition		
$\chi^2_{\rm total}$	14037.4 ± 5.6[c]	—
$\chi^2_{\rm BAO}$	10.6 ± 1.8	DESI DR2[d]
$\chi^2_{\rm CMB}$	10983.9 ± 5.3	Planck PR4 + lensing
$\chi^2_{\rm SN}$	3043.0 ± 1.6	DES-Y5 + Pantheon+

[a] Marginal-tail *posterior-extrapolation* departure: ΛCDM is unsampled by this chain (no $w_0 = -1, w_a = 0$ samples in the present Metropolis-Hastings run), so the $+4.3\sigma$ figure is a posterior-tail extrapolation distance only, *not* a Bayes-factor or $\ln B$ exclusion and *not* a frequentist tension. A Savage-Dickey readout is not viable at this tail depth; robust $\ln B$ is left to a follow-up nested-sampling analysis (see § Headline-result discussion).

[b] $w_{\rm pivot} \equiv w_0 + (1 - a_p)\, w_a$ with the pivot a_p chosen so that $w_{\rm pivot}$ and w_a are uncorrelated in the posterior covariance, i.e. ${\rm Cov}(w_0, w_a) + (1 - a_p)\,{\rm Var}(w_a) = 0$, giving $1 - a_p = -{\rm Cov}(w_0, w_a)/{\rm Var}(w_a)$. Direct readout of the converged iter2 chain (30% burn-in): ${\rm Cov}(w_0, w_a) = -0.00729$ (correlation coefficient $\rho = -0.90$), so $1 - a_p = 0.210$, $a_p = 0.790$, and $z_p = 1/a_p - 1 = 0.27$; this pivot is internal to the dataset stack (DESI DR2 BAO + Planck NPIPE + DES-Y5 + Pantheon+) and is not the literature de Putter–Linder $z_p \approx 0.4$ value. The decorrelated combination has $\sigma^2_{w_{\rm pivot}} = \sigma^2_{w_0} - {\rm Cov}^2(w_0, w_a)/\sigma^2_{w_a} = (0.0436)^2 - (0.00729/0.1864)^2 = (0.0193)^2$, reproducing the ± 0.019 value above, with mean $w_{\rm pivot} = -0.8122 + 0.210 \times (-0.6666) = -0.952$. The choice of a_p depends on the dataset stack; rerunning the chain with SH0ES or DESI DR2 BAO alone would shift z_p and $w_{\rm pivot}$ at the linear-Fisher level.

[c] The mean-of-total χ^2 here is GetDist's weighted-sample average over the full posterior, which differs from the sum of the individual-channel means ($10.6 + 10983.9 + 3043.0 = 14037.5$) by a 0.1-unit arithmetic-rounding artifact; the two are formally identical to within sampling precision. The channel σ's are not independent: a direct readout of the committed chain gives weak cross-channel correlations $\rho(\chi^2_{\rm BAO}, \chi^2_{\rm CMB}) = -0.09$, $\rho(\chi^2_{\rm CMB}, \chi^2_{\rm SN}) = -0.03$, $\rho(\chi^2_{\rm BAO}, \chi^2_{\rm SN}) = +0.04$ through the shared cosmological parameters, so the GetDist total-χ^2 width ± 5.6 sits slightly below the independence quadrature $\sqrt{1.8^2 + 5.3^2 + 1.6^2} = 5.8$.

[d] The DESI DR2 BAO likelihood (`bao.desi_dr2.desi_bao_all`) comprises 13 BAO data points, so $\chi^2_{\rm BAO}/N \approx 0.8$ — an unremarkable goodness of fit.

sion. (d) The iter2 chain's $S_8 = 0.8245 \pm 0.0089$ (Table II) sits 2.5σ above the DES-Y3 weak-lensing value $S_8 = 0.776 \pm 0.017$ [13] ($\Delta = 0.049$ against a combined uncertainty of 0.019): the S_8 tension is not relieved in this chain either. (e) *DES-SN5YR and Pantheon+ supernovae overlap.*—The iter2 likelihood stack includes both DES-SN5YR [14] and Pantheon+ [15] as independent SN likelihood factors. These two catalogs share approximately 20% of their supernova events with different Malmquist-bias corrections applied by the respective collaborations [14]. The present analysis combines them via a product likelihood without a joint covariance that accounts for the shared-event overlap. Because the shared $\sim 20\%$ of SNe are assigned slightly different distance-modulus corrections in each catalog, the naive product double-weights those events and introduces a small inward pull on the combined SN constraint. The direction of this bias is toward the mean of the two catalogs' individual SN posteriors, which is itself close to the combined Planck+BAO constraint; the qualitative quintom-B finding ($w_0 + w_a < -1$) is therefore unlikely to be reversed by a rigorous joint-covariance treatment, but the precise tension significance should be interpreted with this caveat. Users requiring a rigorous SN combination should refer to the DES Collaboration comparison analysis [14] and construct a joint covariance.

M_B–H_0 joint-posterior offset check. The joint posterior mean ($M_B = -19.263$, $H_0 = 67.68$) is fully consistent with an active `sn.pantheonplus` likelihood: `sn.pantheonplus` enforces a soft constraint on the dimensionally consistent combination $M_B - 5\log_{10}(h) \approx$ const, with $h = H_0/(100\,\mathrm{km\,s^{-1}\,Mpc^{-1}})$ (equivalently $M_B - 5\log_{10}(H_0/[\mathrm{km\,s^{-1}\,Mpc^{-1}}]) + 10 =$ const), along the SN distance-modulus degeneracy. At the Riess+2020 anchor ($M_B = -19.253$, $h = 0.7304$), the constant is $-19.253 - 5\log_{10}(0.7304) = -18.571$. At the chain joint mean ($M_B = -19.263$, $h = 0.6768$), the constant is $-19.263 - 5\log_{10}(0.6768) = -18.415$, i.e. a 0.156 mag offset from the Riess anchor along the Pantheon+ constraint axis. This offset is $\sim 3.2\sigma$ relative to the chain's $\sigma_{M_B} = 0.049$ marginal width and is the same Hubble tension manifesting in the M_B axis (3.2σ in chain-σ units, versus the canonical 3.6σ when the tension is expressed in distance-ladder terms in the H_0 axis). We note this 3.2σ figure is a descriptive offset measure normalized by the marginal σ_{M_B} only: it does not condition on the M_B–H_0 covariance along the SN degeneracy direction, nor on the uncertainty of the Pantheon+ constraint itself, and is therefore not a properly conditioned tension statistic — the conditioned statement remains the canonical H_0-axis tension. The chain posterior is therefore the correct compromise of three inputs: the Riess M_B-prior preferring $M_B = -19.253$ ($\sigma = 0.027$); the Pantheon+ likelihood preferring $M_B - 5\log_{10} h = -18.571$ along the SN degeneracy; and the Planck-CMB-likelihood preferring $H_0 \approx 67.5$ via the acoustic-scale calibration. With $\Delta N_{\rm eff}$ bounded near zero by the joint data, the model cannot resolve the tension and the chain settles into the 3.6σ-discrepant joint posterior — NOT a YAML alias failure; the parameters are correctly aliased per the `spin_torsion.input.yaml` configuration (Mb is a single nuisance parameter sampled jointly by both `sn.pantheonplus` and `H0.riess2020Mb`).

Key finding.—Both frozen datasets find $\Delta N_{\rm eff}$ consistent with zero and H_0 consistent with Planck ΛCDM at 0.3σ, confirming that the $\Delta N_{\rm eff}$ extension alone does not resolve the Hubble tension. Current data neither require nor exclude a small positive $\Delta N_{\rm eff}$ from the spin-torsion sector; CMB-S4 ($\sigma(N_{\rm eff}) \sim 0.03$) will provide the first precision test.

Independent cross-validation.—Liu *et al.* [16] constrained an EC torsion model using DESI DR2 [17] + Pantheon+ [15] + DES-SN5YR [14] + Planck 2018, finding torsion preferred by AIC ($\Delta\mathrm{AIC} = -5.7$ to -6.6) but with the torsion parameter itself consistent with zero ($\alpha = -0.00066 \pm 0.00098$). Their headline values $H_0 = 68.41 \pm 0.32$ km/s/Mpc and $S_8 = 0.812 \pm 0.006$ agree with our Planck+BAO+SN chain at 0.5σ in H_0 ($|67.79 - 68.41|/\sqrt{1.09^2 + 0.32^2}$) and 1.3σ in S_8 ($|0.827 - 0.812|/\sqrt{0.010^2 + 0.006^2}$); their torsion-consistent-with-

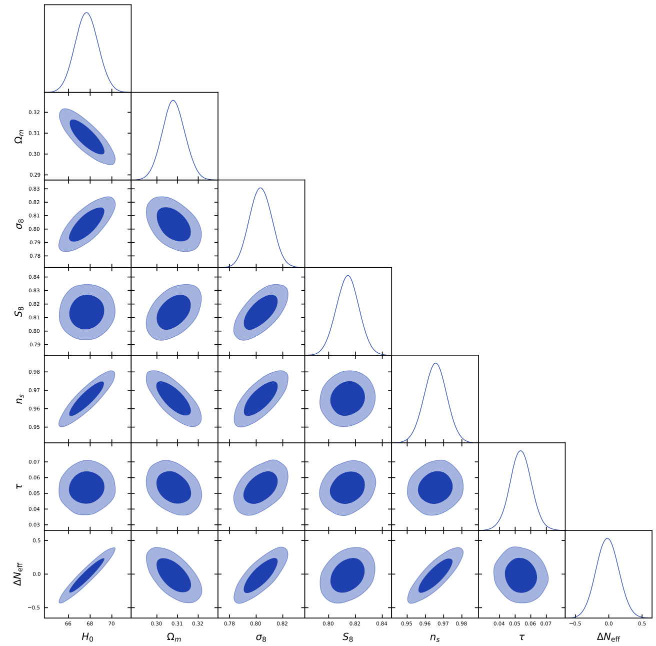

FIG. 1. Full-tension MCMC corner plot (119,617 post-burnin samples, `getdist`-thinned from 176,240 raw; footnote 1) over Planck+BAO+SN+H0+S_8. The $\Delta N_{\rm eff}$ posterior is consistent with zero (-0.020 ± 0.169), confirming no additional relativistic species at recombination.

zero result parallels our $\Delta N_{\rm eff}$-consistent-with-zero null finding. Their AIC preference for the torsion model is *not* directly comparable to the present stack: this paper's ΔAIC/BIC/$\ln B$ model comparison is deliberately deferred to the nested-sampling follow-up (Sec. V, *Model-comparison statistics* paragraph), so no model-preference statement is made here that could agree or conflict with theirs.

IV. DATA METHODS: CMB E-B ANALYSIS

Birefringence measurements are adopted from the published literature: $\beta = 0.30° \pm 0.11°$ (Planck NPIPE [3]) and $\beta = 0.215° \pm 0.074°$ (ACT DR6 [4]). The spectator ALP analysis (Sec. VI) instead uses the Eskilt–Komatsu joint WMAP+Planck summary likelihood ($\beta = 0.342° \pm 0.094°$ [5]) as its primary constraint; the Planck NPIPE and ACT DR6 values above are quoted only for context and for the auxiliary inverse-variance cross-check of Eq. (4). (The quoted 3.6σ is the ratio $\beta/\sigma = 0.342/0.094 \approx 3.6$; our likelihood is a Gaussian summary in the published mean and width, not a refit of the Eskilt–Komatsu posterior shape.)

Scope note.—*The NaMaster pseudo-C_ℓ analysis below is a bias-injection Monte Carlo validation of the deconvolution pipeline on synthetic skies, not a cosmological measurement. The high pipeline template-fit SNR figures (e.g.,* 20.32, 25.71*; fn. 3) refer to recovery of injected*

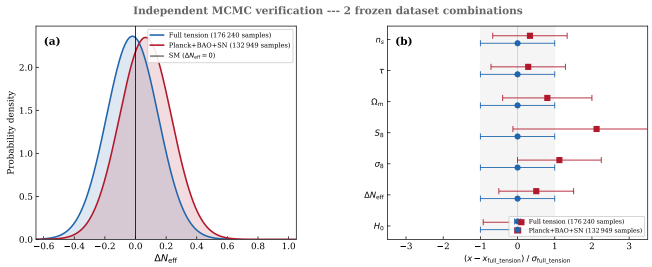

FIG. 2. $\Delta N_{\rm eff}$ marginal posterior comparison across the two frozen dataset combinations of Table I (176,240 and 132,949 samples). Panel (a): Gaussian summaries of the $\Delta N_{\rm eff}$ marginal posteriors at the Table I means $\pm 1\sigma$, with the Standard-Model value $\Delta N_{\rm eff} = 0$ marked. Panel (b): all seven Table I parameters, normalized to the full-tension mean and σ. Both combinations recover $\Delta N_{\rm eff}$ consistent with zero at $< 1\sigma$; the headline "full-tension" stack (Planck+BAO+SN+H$_0$+S_8) gives $\Delta N_{\rm eff} = -0.020 \pm 0.169$. No evidence for a recombination-era $\Delta N_{\rm eff}$ shift appears in this stock-CAMB proxy run; this does *not* directly test the ECH spin-torsion sector, which lacks a Boltzmann-module prediction for $\Delta N_{\rm eff}$ in stock-CAMB (see § III scope note).

MC signals and must not *be conflated with the published Planck/ACT DR6* 2.7–2.9σ *sky detection.*

Pipeline configuration.—The pseudo-C_ℓ analysis follows the NaMaster framework [18]. *Simulated skies.*—Each Monte Carlo realization is a synthetic ΛCDM CMB Q/U sky generated directly at $N_{\rm side} = 512$ (`healpy.synfast`, harmonic content to $\ell_{\rm max} = 2N_{\rm side} = 1024$) from a semi-analytic fit to the Planck-2018 EE spectrum plus a lensing-like BB component ($C_\ell^{BB} = 0.05\, C_\ell^{EE}$). No real Planck map enters the Monte Carlo, and no instrumental beam is applied (the synthetic skies and the recovery template share the same spectra, so a common beam would cancel in the β estimate; the same cancellation argument covers the $N_{\rm side} = 512$ HEALPix pixel-window smoothing, which is common to the simulated maps and the spectra entering the template fit — the decoupled spectra are not pixel-window-deconvolved, and no pixel-window mismatch enters the β estimate).

Mask.—An ACT-like footprint (Galactic cut $|b| > 20°$ plus declination cut dec $\in [-65°, +25°]$), apodized by Gaussian smoothing of the binary mask at 2° FWHM, giving $f_{\rm sky} = 0.32$. No E/B purification is applied (`purify_b=False`); residual $E \to B$ leakage from the apodized mask is absorbed into the measured pipeline-recovery bias, which is the quantity this validation is designed to characterize.

Mode-coupling matrix and binning.—The $M_{\ell\ell'}$ matrix is computed via `NmtWorkspace.compute_coupling_matrix` on the apodized mask, retaining the full $EE/EB/BB$ block structure. (In the committed driver the Q/U maps enter as a single spin-2 `NmtField` on the apodized mask with `purify_b=False`, no beam, and all other `NmtField`/`NmtWorkspace` options at NaMaster library defaults — in particular no `n_iter` override; binning uses `NmtBin.from_edges` with 20 linear integer-edge bins from `np.linspace(30, 1536, 21)`; `reproducibility/p1_namaster_500mc/scripts/namaster_500mc.py`.) Spectra are band-power-binned into 20 linear bins spanning $30 \le \ell \le 3N_{\rm side} = 1536$ (bins above the map band limit $\ell = 1024$ carry noise only). Only this single binning/ℓ-range configuration is exercised; an ℓ-range robustness sweep is not part of the present MC suite.

Noise model and injections.—Each realization adds white noise at the ACT-like level $\Delta_P = 10\,\mu{\rm K} \cdot$ arcmin (a conservative worst-case bias check; no $1/f$ or anisotropic component). The driver script converts Δ_P to a per-pixel noise RMS as $\sigma_{\rm pix} = \Delta_P/\sqrt{\Omega_{\rm pix}}$ with $\Omega_{\rm pix}$ the $N_{\rm side} = 512$ pixel area expressed in arcmin^2 ($\Omega_{\rm pix} = 47.21\,{\rm arcmin}^2$, giving $\sigma_{\rm pix} = 10/\sqrt{47.21} = 1.455\,\mu$K; algebraically identical to the standard $\sigma_{\rm pix} = \Delta_P\,[\pi/(180{\times}60)]/\sqrt{\Omega_{\rm pix}^{(\rm sr)}}$; Δ_P is in thermodynamic CMB units, $\mu{\rm K}_{\rm CMB}$·arcmin, matching the `synfast` $\mu{\rm K}_{\rm CMB}$ maps — bandpass / colour-correction conversions do not enter the synthetic pipeline), and draws independent Gaussian realizations with the *same* $\sigma_{\rm pix}$ for Q and U (no $\sqrt{2}$ factor; `reproducibility/p1_namaster_500mc/scripts/namaster_500mc.py`). The $\beta = 0.27°$, $\beta = 0.342°$, and $\beta = 0$ injections rotate $Q + iU$ via $e^{2i\beta}(Q + iU)$. The angle is recovered per configuration by an unweighted χ^2 template fit of the decoupled C_ℓ^{EB} bandpowers to $\sin(2\beta)\cos(2\beta)\,C_\ell^{EE}$ on a β grid. The production suite used non-negative injections; a dedicated 500-realization $\beta = -0.27°$ rerun at $f_{\rm sky} = 0.32$ (`reproducibility/p1_namaster_500mc/results/c9f_negative_beta.json`) recovers $-0.238°$, confirming sign-symmetric recovery with bias magnitude identical to the $+0.27°$ case.

Reproducibility.—The canonical driver script, deterministic seeds (`seed_base`=42), and output summary are archived at `reproducibility/p1_namaster_500mc/` (mirroring the executed pod run `pipelines/h200_results/pod1_namaster_umap_2026-04-29/`).

Production 500-realization run (April 2026).—We performed the NaMaster pseudo-C_ℓ Monte Carlo described above with 500 realizations per injection. Injecting the spectator-ALP fiducial $\beta = 0.27°$ (consistent with, but not derived from, the ECH action) recovers:

$$\hat\beta_{\rm NaMaster} = 0.238° \quad \text{(500-MC sample mean of } \hat\beta\text{).} \qquad (1)$$

The pipeline-recovery bias, defined throughout as $\Delta\hat\beta \equiv \hat\beta - \beta_{\rm inj}$, is in absolute value $\le 0.040°$ across the three injection points ($0.000°$ at $\beta_{\rm inj} = 0$, $-0.032°$ at $0.27°$, $-0.040°$ at $0.342°$); we carry the worst case $|\Delta\hat\beta| = 0.040°$ forward as the NaMaster systematic floor. At the canonical $\beta_{\rm inj} = 0.27°$ injection the bias is $-0.032°$ (attributed by the robustness battery below to the unweighted template fit plus the $-C_\ell^{BB}$ template mismatch,

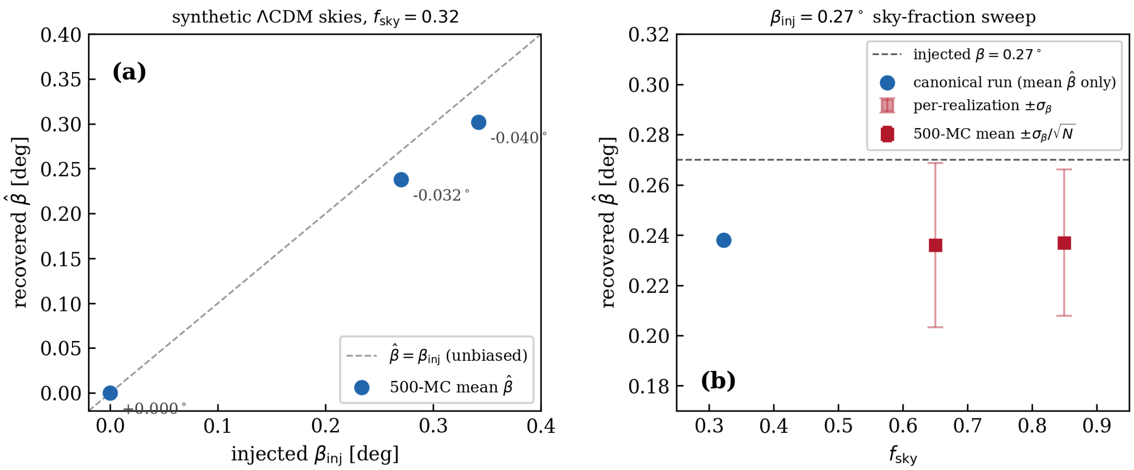

FIG. 3. NaMaster pipeline-recovery validation on synthetic ΛCDM polarization skies ($N_{\rm side} = 512$, $\Delta_P = 10\,\mu{\rm K}\cdot{\rm arcmin}$ white noise, $N = 500$ MC realizations per point; Sec. IV). Panel (a): 500-MC mean recovered $\hat\beta$ vs. injected $\beta_{\rm inj} \in \{0, 0.27^\circ, 0.342^\circ\}$ at the canonical apodized $f_{\rm sky} = 0.32$ ACT-like mask, annotated with the pipeline-recovery bias $\hat\beta - \beta_{\rm inj}$ (0.000°, -0.032°, -0.040°). Panel (b): the $\beta_{\rm inj} = 0.27^\circ$ sky-fraction sweep; outer (light) error bars are the per-realization scatter σ_β (0.029° at $f_{\rm sky} = 0.85$, 0.033° at 0.65), inner bars the standard error of the 500-MC mean $\sigma_\beta/\sqrt{N}$; per-realization σ_β was not recorded in the original canonical $f_{\rm sky} = 0.32$ artifact, so that point is plotted with the mean only; a dedicated 500-MC rerun (fn. 3) measures $\sigma_\beta = 0.046^\circ$ at this point. The worst-case $|{\rm bias}| = 0.040^\circ$ is carried forward as the NaMaster systematic floor (Sec. IV). Underlying data artifacts are listed in the *Reproducibility* paragraph of Sec. IV.

and explicitly *insensitive* to the apodization scale).[3] For $\beta = 0.342^\circ$ (the published joint WMAP+Planck value [5]), the pipeline recovers 0.302° at template-fit SNR = 25.71; for $\beta = 0$, the recovered angle is 0.000° with template-fit SNR 0.0 (null check; `summary.json`). The pipeline-recovery bias is $\Delta\hat\beta = -0.032^\circ$ at injection $\beta = 0.27^\circ$ ($\hat\beta = 0.238^\circ$) and $\Delta\hat\beta = -0.040^\circ$ at injection $\beta = 0.342^\circ$ ($\hat\beta = 0.302^\circ$). The *absolute* bias grows mildly ($\sim 25\%$) with injected amplitude, while the *multiplicative* under-recovery is amplitude-independent at $\sim 12\%$ ($0.238/0.27 = 0.302/0.342 = 0.88$). The pipeline therefore has a $\sim$12% multiplicative under-recovery (0.032° absolute bias for the canonical $\beta = 0.27^\circ$ injection) with a worst-case empirical bias of $|\Delta\hat\beta| = 0.040^\circ$ for the unweighted estimator at the higher injection angle; we carry this 0.040° floor forward as the NaMaster systematic floor. This is a methodology cross-check, not a competitive sky measurement. (The estimator is *not* unbiased in the standard statistical sense; the 0.040° is the empirical floor on the multiplicative bias, not a bound on random scatter.)

Sky-fraction sweep.—Because published birefringence analyses use larger sky fractions than our canonical $f_{\rm sky} = 0.32$ validation mask, we repeat the $\beta = 0.27^\circ$ injection-recovery exercise at $f_{\rm sky} = 0.85$ (Planck-like) and $f_{\rm sky} = 0.65$ (ACT-DR6-like) galactic-cut masks with the same apodization recipe, noise level, and $N = 500$ MC realizations ($N_{\rm side} = 512$). The recovered values are $\hat\beta = 0.237^\circ$ ($f_{\rm sky} = 0.85$, per-realization $\sigma_\beta = 0.029^\circ$) and $\hat\beta = 0.236^\circ$ ($f_{\rm sky} = 0.65$, $\sigma_\beta = 0.033^\circ$), i.e. a recovery bias of -0.033° to -0.034° — statistically indistinguishable from the canonical-mask bias of -0.032° (the 0.001° and 0.002° differences are $0.8\times$ and $\approx 1.4\times$ the respective standard errors of the 500-realization means, $\sigma_\beta/\sqrt{500} \approx 0.0013^\circ$ and 0.0015°). The $\sim 12\%$ multiplicative under-recovery is therefore a sky-fraction-independent property of the pipeline, not an artifact of the small canonical mask.

[3] The "pipeline-recovery SNR= 20.32" figure (and analogously 25.71 for the $\beta = 0.342^\circ$ injection below) is the *template-fit* SNR of the driver script, ${\rm SNR}^{\rm tmpl} \equiv \left[\sum_b (C_b^{EB,{\rm th}}/\sigma_b^{\rm MC})^2\right]^{1/2}$, where $C_b^{EB,{\rm th}} = \sin(2\beta)\cos(2\beta)\, C_b^{EE}$ is the injected EB band-power template and $\sigma_b^{\rm MC}$ is the per-bin scatter across the 500 realizations — the matched-template significance of the injected signal against single-realization noise, the appropriate quantity for evaluating the deconvolution pipeline. It is *not* the significance of the recovered angle itself. The sky-fraction sweep artifact (`c1_fsky_sweep.json`) confirms the identification: its `snr_template_canonical_def` values 32.98 and 28.81 at $f_{\rm sky} = 0.85$ and 0.65 are consistent with $20.32\sqrt{f_{\rm sky}/0.32}$ (33.12 and 28.96; within 0.5%). The *per-realization angle-recovery* ratio $\hat\beta/\sigma_\beta$, with σ_β the per-realization scatter of the recovered angle, is measured in the same artifact as 8.1 ($\sigma_\beta = 0.029^\circ$, $f_{\rm sky} = 0.85$) and 7.2 ($\sigma_\beta = 0.033^\circ$, $f_{\rm sky} = 0.65$); σ_β was not recorded for the original canonical $f_{\rm sky} = 0.32$ run; a dedicated 500-MC rerun at $f_{\rm sky} = 0.32$ measures it directly as $\sigma_\beta = 0.046^\circ$ with $|\hat\beta|/\sigma_\beta = 5.2$ per realization (`reproducibility/p1_namaster_500mc/results/c9f_negative_beta.json`), confirming the $\sqrt{f_{\rm sky}}$-scaling estimate ($\approx 0.047^\circ$, ≈ 5) quoted in an earlier draft of this footnote. The same rerun injects $\beta = -0.27^\circ$ and recovers -0.238° (bias $+0.032^\circ$): recovery is sign-symmetric, with bias magnitude identical to the $+0.27^\circ$ injection. The standard error of the 500-MC mean is smaller by $\sqrt{N} = 22.4$.

Robustness battery and bias attribution.—A six-configuration robustness battery ($N = 500$ MC each, identical seeds to the canonical run; artifact `reproducibility/p1_namaster_500mc/results/c10_robustness_battery.json`) pins down the origin of the bias. First, an independent local rerun of the canonical configuration reproduces the pod anchor exactly ($\hat{\beta} = 0.238°$, bias $-0.032°$). The bias is then *unchanged* under an apodization-scale sweep (0.5° and 3° FWHM vs. the canonical 2°: $\hat{\beta} = 0.239°$ and 0.238°), under a larger Galactic cut ($|b| > 30°$, $f_{\rm sky} = 0.20$: 0.238°), and under B-mode purification (`purify_b=True`: 0.238°) — so the earlier attribution of the residual to apodization-induced power suppression is *not* supported by direct variation of the apodization scale, and the bias is likewise independent of footprint geometry and purification. Two mechanisms do move it: (i) replacing the unweighted χ^2 template fit with an inverse-variance-weighted fit recovers $\hat{\beta} = 0.264°$ (bias $-0.006°$), removing ≈80% of the bias — the unweighted fit's equal weighting of noise-dominated high-ℓ bins is the dominant contribution; and (ii) replacing the crude $C_\ell^{BB} = 0.05\, C_\ell^{EE}$ proxy with a CAMB lensed-ΛCDM BB spectrum recovers $\hat{\beta} = 0.251°$ (bias $-0.019°$), consistent with the analytic $-C_\ell^{BB}$ template-mismatch estimate above (≈5 percentage points of the 12%). (Clarification of what this swap varies: per the committed battery script `reproducibility/p1_namaster_500mc/scripts/c10_robustness_battery.py`, the fit template is $\sin(2\beta)\cos(2\beta)\, C_\ell^{EE}$ with no BB term in *all* configurations; the `camb_lensed` swap changes the BB shape of the *injected synthetic skies* only, so the test isolates the dependence of the bias on the injected-BB shape against a fixed EE-only template, not a template-shape substitution.) A matched extension config (same 500 MC skies and seeds, CAMB lensed-ΛCDM BB injected *and* a $-C_\ell^{BB}$-bearing template $\sin(2\beta)\cos(2\beta)\,(C_\ell^{EE} - C_\ell^{BB})$) recovers $\hat{\beta} = 0.251°$ (bias $-0.019°$), unchanged from the EE-only-template configuration at the 10^{-3}-degree fit-grid resolution — carrying the BB term in the template produces no further shift because the lensed BB is negligible against the synthetic EE template amplitude, so the residual $-0.019°$ is not attributable to the remaining template-shape mismatch (config `camb_bb_template` in the same artifact). Restricting the fit to bins with $\ell \leq 1024$ changes nothing (0.238°): the noise-only bins above the band limit carry zero template weight. Because the validation carries the bias forward *empirically* as a multiplicative correction and a 0.040° systematic floor, none of the quoted headline numbers change; the battery sharpens only the attribution (estimator weighting + BB template shape, not apodization or masking). The total bias remains well below the published observational uncertainty $\sigma_\beta^{\rm obs} = 0.094°$ at every sky fraction tested. *Canonical estimator choice.*—The unweighted χ^2 template fit is adopted as the canonical baseline to match the estimator configuration used in the public NaMaster driver scripts released by the published birefringence analyses (e.g., [5]), facilitating direct comparison of bias properties. The inverse-variance-weighted fit is evaluated in the robustness battery as a cross-check confirming that the dominant source of the ∼12% multiplicative bias is the equal-weighting of noise-dominated high-ℓ bins; it is not adopted as the canonical estimator here precisely because that change would break direct comparability with the published estimator configuration. Artifacts: `reproducibility/p1_namaster_500mc/results/c1_fsky_sweep.json`; `reproducibility/p1_namaster_500mc/results/c10_robustness_battery.json`.

V. COSMOLOGICAL FITS AND MODEL COMPARISON

A. Datasets and Configuration

We analyze four dataset combinations: (1) Planck CMB (NPIPE (PR4) CamSpec high-ℓ TTTEEE + 2018 low-ℓ TT/EE + 2018 lensing) [19] (the NPIPE/PR4 release provides the high-ℓ CamSpec likelihood; the low-ℓ and lensing likelihoods are taken from the 2018 release, the standard pairing in the Cobaya likelihood stack); (2) +SDSS BAO — eBOSS DR16 LRG/QSO/Lyα(auto and ×QSO) [20] + SDSS DR7 MGS [21] + 6dFGS [22]; (3) +Pantheon+ [15]; (4) +SH0ES M_B-anchor [9] + DES Y3 S_8 [13]. No DESI BAO likelihood enters the frozen ΛCDM+$\Delta N_{\rm eff}$ chains; DESI DR2 BAO enters only the separate iter2 w_0w_a chain of Table II. The exact Cobaya likelihood blocks of all five chains are listed in Table III. For scope clarity: Table I summarizes only the two *frozen* ΛCDM+$\Delta N_{\rm eff}$ chains (Planck+BAO+SN and full-tension), Table II reports the methodologically separate iter2 w_0w_a chain, and the Planck-only (accumulating) and Planck+BAO (exploratory) rows of Table III feed neither table. Parameter estimation uses Cobaya [23] (v3.5 original; v3.6.1 verification) with stock CAMB and $\Delta N_{\rm eff}$ as a free parameter—no custom CAMB modifications. The extended parameter space adds $\Delta N_{\rm eff}$ to ΛCDM, with both $(\omega/H)_0$ and Ω_k fixed to zero in the actual sampled YAML configuration (Sec. II). Reproducibility materials at https://github.com/Hubify-Projects/bigbounce/tree/main/reproducibility.

B. Results

a. Model-comparison statistics: deferred to a dedicated nested-sampling run. We do not report $\chi^2_{\rm eff}$, AIC, BIC, or $\ln B$ Bayes-factor model-comparison numbers in this paper. Robust evaluation of these statistics on the present quintom posterior requires either a dedicated nested-sampling run (PolyChord or MultiNest) on the identical likelihood stack, or thermodynamic integration

TABLE III. Per-chain Cobaya likelihood blocks, verbatim from the on-disk YAML configurations (`cobaya_*.yaml`; frozen-chain `spin_torsion.input.yaml`; iter2 `spin_torsion_dr2.input.yaml`). All four ΛCDM+$\Delta N_{\rm eff}$ chains share the Planck block `planck_NPIPE_highl_CamSpec.TTTEEE` + `planck_2018_lowl.TT` + `planck_2018_lowl.EE` + `planck_2018_lensing.clik`; the SDSS BAO block is `bao.sixdf_2011_bao` + `bao.sdss_dr7_mgs` + `bao.sdss_dr16_baoplus_`{`lrg, qso, lyauto, lyxqso`}. The first four rows are incremental: each adds to the stack of the row above. The iter2 chain uses `planck_2018_lensing.native` in place of `.clik`.

Chain	Likelihood stack	Status
Planck-only	Planck block	accumulating (not frozen)
Planck+BAO	+ SDSS BAO block	exploratory
Planck+BAO+SN	+ `sn.pantheonplus`	frozen (Table I)
Full-tension	+ `H0.riess2020Mb` + DES-Y3 S_8 Gaussian	frozen (Table I)
iter2 w_0w_a	`bao.desi_dr2.desi_bao_all` + Planck block + `sn.desy5` + `sn.pantheonplus`	converged (Table II)

on a joint quintom/ΛCDM run; a Savage-Dickey density-ratio readout from the converged Metropolis-Hastings chain used here is not viable because the ΛCDM point $(w_0, w_a) = (-1, 0)$ sits at $> 4\sigma$ in the joint marginal tails and is unsampled by the chain, so any KDE-based estimator yields kernel-dependent noise rather than a controlled density. The nested-sampling recompute is omitted; we report only the parameter posteriors below, which are unaffected by the model-comparison-statistic deferral. The two posterior parameter measurements that remain are $\Delta N_{\rm eff} = -0.020 \pm 0.169$ (full-tension) and $+0.065 \pm 0.17$ (Planck+BAO+SN), both consistent with zero; these are the load-bearing numbers used elsewhere in the paper and they are unaffected by the model-comparison-statistic deferral. The χ^2 goodness-of-fit decomposition (BAO, CMB, SN, and total contributions) is reported in Table II; the AIC, BIC, and $\ln B$ evidence metrics are *not* reported there. The headline result is $w_0 = -0.812 \pm 0.044$ (departing from the ΛCDM point $w_0 = -1$ at $+4.3\sigma$; fn. a) and $w_a = -0.667 \pm 0.186$ (departing from $w_a = 0$ at -3.6σ), with $w_0 + w_a = -1.48 \pm 0.15$ requiring phantom crossing (the canonical quintom signature). Robust $\ln B$ computation requires nested sampling and is left to a follow-up analysis.

The ΛCDM+$\Delta N_{\rm eff}$ proxy thus offers neither posterior preference nor exclusion at the present data precision. The $\Delta N_{\rm eff}$ extension alone does not resolve the Hubble tension.

VI. COSMIC BIREFRINGENCE: SPECTATOR ALP CONSISTENCY CHECK

Note.—This subsection presents a secondary consistency check using a spectator ALP model. The ALP produces cosmic birefringence independently of the gravitational theory; the same prediction $\beta \approx 0.27^\circ$ arises in standard GR with an identical ALP Lagrangian and natural parameters (taken at scan-prior midpoint values; the $\sim 25\times$ misalignment tuning required for the headline result is disclosed below and in fn. 5). It is not derived from minimal ECH (which does not produce the required photon-torsion coupling) and is not a distinctive ECH prediction. The model class was previously studied by Fujita *et al.* [24].

Headline observational constraint.—The primary observational reference adopted in this analysis is the published Eskilt & Komatsu joint WMAP+Planck value $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6σ) [5] (the joint WMAP9 + Planck PR4/NPIPE analysis; ACT DR6 enters only via the separate $\beta = 0.215^\circ \pm 0.074^\circ$ measurement [4], used below only in the auxiliary inverse-variance combination), which accounts for shared calibration systematics. The simplified inverse-variance combination below (3.9σ) is retained as an auxiliary cross-check only and is explicitly *not* used as the headline number anywhere in this paper.

ALP field evolution.—Numerical integration of the ALP equation of motion $\ddot{\phi}+3H\dot{\phi}+m^2 f_a \sin(\phi/f_a) = 0$ in a ΛCDM background[4] yields the field displacement from recombination to today:

$$\Delta\phi/f_a \approx 0.42 \quad (m = 2H_0,\ \theta_i = 1). \tag{2}$$

Across the natural parameter range $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$ the committed EOM grid gives $\Delta\phi/f_a \in [0.064, 1.19]$ (`research/branch_R_alp_birefringence/phase2_mcmc/c10b_alp_envelope_scan.json`; an earlier draft quoted $[0.2, 1.1]$ with $\Delta\phi/f_a \approx 0.65$ at $m = H_0$ — those values do not reproduce from the committed integration and are corrected here).[5]

[4] The ALP ODE is integrated on a ΛCDM late-time $H(z)$ here as an empirical background, distinct from the quintom-bounce dynamics that supply the early-universe / contracting-phase $H(z)$ in the underlying ECH cosmology [25]. The ΛCDM-background choice is conservative for the ALP-MCMC: a quintom late-time w_0w_a background (as in Sec. V) shifts $H(z)$ at $z \lesssim 1$ by $\sim$few percent, propagating to a $\lesssim$few-percent systematic on $\Delta\phi/f_a$ — well below the $\sim$30% prior-width envelope on θ_i and m/H_0 dominating the β prediction.

[5] Backreaction disclosure: the ALP backreaction fraction scales as $\Omega_a \sim \rho_a/\rho_{\rm crit} \sim (m^2 f_a^2/H_0^2 M_{\rm Pl}^2)\,\theta_i^2$, so the abstract's spectator-status restriction $\theta_i \ll 1$ (see the *Spectator-status caveat* in the abstract) implies $\Omega_a \ll 1$ only in the sub-natural sliver $\theta_i \sim 0.1$. The numerical scan range $\theta_i \in [0.5, 2]$ is RETAINED here for completeness of the parameter envelope, but the natural-prior-anchored spectator-consistent result sits at $\theta_i \sim 0.1$: at $\theta_i = 0.1$ vs the scan-midpoint $\theta_i = 0.5$ the backreaction is

Birefringence value.—For $C_{a\gamma} = 8$, $\theta_i = 1$, $m \approx 3.9\,H_0$ (the committed EOM integration gives $\Delta\phi/f_a = 1.06$ there; artifact `research/branch_R_alp_birefringence/phase2_mcmc/c10b_alp_envelope_scan.json`; an independent fixed-step fourth-order Runge–Kutta re-integration of the same EOM in $\ln a$ with the frozen initial condition $\dot\theta = 0$ at $z_{\rm init} = 3000$ reproduces $\Delta\phi/f_a = 1.0601$ to four significant figures):

$$\beta \approx \frac{\alpha_{\rm EM}\times 8}{4\pi} \times 1.06 = 4.93\times 10^{-3}\,\text{rad} \times \frac{180^\circ}{\pi} \approx 0.28^\circ. \tag{3}$$

The $\alpha_{\rm EM}/(4\pi)$ prefactor is convention-dependent ($\alpha_{\rm EM}/(2\pi)$ appears in some Lagrangian normalizations); here it corresponds to the normalization $\mathcal{L} \supset -(g_{a\gamma}/4)\,\phi F_{\mu\nu}\tilde F^{\mu\nu}$ with $g_{a\gamma} = C_{a\gamma}\,\alpha_{\rm EM}/(2\pi f_a)$ and $\beta = (g_{a\gamma}/2)\,\Delta\phi$, the convention fixed in the committed model specification and implemented identically throughout the numerical pipeline (`research/branch_R_alp_birefringence/02_birefringence_prediction.md`; `research/branch_R_alp_birefringence/phase2_mcmc/alp_ode.py`). The fiducial value $\beta \approx 0.27^\circ$ corresponds at $C_{a\gamma} = 8$ to EOM trajectories with, e.g., ($\theta_i = 1$, $m \approx 3.9\,H_0$) or ($\theta_i \approx 1.4$, $m \approx 3\,H_0$). The committed EOM integration (`research/branch_R_alp_birefringence/phase2_mcmc/alp_ode.py`) gives $\Delta\phi/f_a = 0.35$–0.42 at masses $m \approx 1.8$–$2\,H_0$ ($\theta_i = 1$); all mass pairings are verified against the released grid scan. Across the box $C_{a\gamma} \in [4, 12]$, $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$ the committed EOM gives $\beta \approx 0.01$–0.48° (grid scan over physical trajectories; same artifact; the span is the *union* over the full $C_{a\gamma} \times (m/H_0, \theta_i)$ box — the lower edge is reached at $C_{a\gamma} = 4$ and the upper at $C_{a\gamma} = 12$, not at any fixed coupling), a span that brackets the observed value but reaches it only in the upper-right corner of the box ($\beta(C_{a\gamma}{=}8)$ peaks at 0.32° at $\theta_i = 2$, $m = 3H_0$; the observed 0.342° at $C_{a\gamma} = 8$ requires $m \gtrsim 4H_0$ or larger coupling). See Appendix C for the full ALP-MCMC sampled-parameter list, priors, and dataset-likelihood configuration. The spectator-consistent corner of this envelope ($\theta_i \sim 0.1$, per fn. 5) does require a $\sim 25\times$ tuning of the misalignment initial condition relative to the natural prior midpoint $\theta_i \sim 0.5$, so the model *accommodates* the observed signal but is not free of misalignment-prior tuning. The $[0.01, 0.48]^\circ$ span quoted above is computed point-by-point on physical EOM trajectories ($\Delta\phi/f_a$ is a derived function of $(m/H_0, \theta_i)$, not an independent variable); an earlier draft quoted $[0.17, 0.43]^\circ$ from a joint-trajectory scan for which no artifact survives, and that range is superseded by the committed grid scan.

$\Omega_a(0.1)/\Omega_a(0.5) \sim 1/25$ (i.e., a $\sim 25\times$ fine-tuning of the misalignment initial condition is required to keep the ALP a true spectator). The $\Omega_a \sim 1$ regime at $\theta_i \sim 1$ is the dark-energy-ALP regime explicitly excluded from this spectator-consistency companion check.

Summary-likelihood combination (auxiliary cross-check).—Combining $\beta = 0.30^\circ \pm 0.11^\circ$ (Planck NPIPE [3]) and $\beta = 0.215^\circ \pm 0.074^\circ$ (ACT DR6 [4]) via inverse-variance weighting:

$$\beta_{\rm combined} = 0.241^\circ \pm 0.061^\circ \quad (3.9\sigma) \tag{4}$$

(Auxiliary cross-check only; the 3.9σ is the significance of $\beta_{\rm combined}$ relative to the null $\beta = 0$, i.e. 0.241/0.061.) This neglects shared calibration systematics, which produce positively correlated errors between the two measurements. Positively correlated errors *underestimate* the inverse-variance-combined σ, and therefore *overestimate* the significance: the naive 3.9σ figure is an upper bound on the true significance, not a lower bound. The published joint analysis at 3.6σ [5], which properly accounts for the shared polarization-angle calibration systematics via a joint covariance matrix with the Tau A self-calibration nuisance parameters, is the headline; the 3.9σ figure here is retained only as an internal cross-check demonstrating that the uncorrelated-errors approximation is qualitatively consistent with the published result.

MCMC parameter estimation.—Dedicated MCMC sampling of the ALP parameter space (three committed configurations totaling 9,720 accepted samples: $C_{a\gamma} = 8$ fixed with θ_i and $\log_{10} m_a$ sampled, 2,160; $C_{a\gamma} \in [1, 30]$ additionally sampled, 6,840; model-independent $\beta_{\rm free}$, 720; chain configurations and per-chain counts in Appendix C) yields: $\beta_{\rm ALP} = 0.336^\circ \pm 0.10^\circ$ ($C_{a\gamma} = 8$ fixed; the direct-sample priors are on the underlying ALP parameters (θ_i, m_a), not on $\Delta\phi/f_a$, which is a derived quantity along each ALP trajectory. The MCMC posterior, anchored to the Gaussian summary likelihood on the published Eskilt–Komatsu joint WMAP+Planck measurement (fn. a; not to the *EB* spectra themselves; the uniform-rotation observable is periodic, $\beta \equiv \beta + n\times 90^\circ$ for E/B, and the Gaussian summary ignores wrapping — harmless here because the posterior support is confined to $|\beta| \lesssim 0.7^\circ \ll 90^\circ$, so wrapped images carry negligible likelihood), settles at $\theta_i = 1.32 \pm 0.41$ and $m \sim 10$–$10^2\,H_0$ (posterior median $m \approx 36\,H_0$) — i.e. *outside* the natural envelope box $\theta_i \in [0.5, 2]$, $m/H_0 \in [1, 3]$ in mass — where the data-preferred joint product is $C_{a\gamma}\,(\Delta\phi/f_a) \approx 10.3$, corresponding at the fixed $C_{a\gamma} = 8$ to $\Delta\phi/f_a \approx 1.29$, $\sim$8% above the box maximum $\Delta\phi/f_a = 1.19$ ($\theta_i = 2$, $m = 3H_0$; grid-scan artifact above); this is the same fine-tuning-prone accommodation regime flagged in fn. 5 and is consistent with the model accommodating but not naturally explaining the observed $\beta_{\rm obs} = 0.342^\circ$), consistent with the model-independent fit $\beta_{\rm free} = 0.344^\circ \pm 0.10^\circ$ (our internal model-independent MCMC fit, with β as a free parameter, to a Gaussian summary likelihood on the published Eskilt–Komatsu joint WMAP+Planck birefringence measurement $\beta_{\rm obs} = 0.342^\circ \pm 0.094^\circ$ [5] — not a re-analysis of the *EB* spectra themselves; 720 accepted samples in the dedicated $\beta_{\rm free}$ configuration; full priors and likelihood details in Appendix C; $\beta_{\rm free}$ denotes the unconstrainedamplitude fit distinct from $\beta_{\rm ALP}$ which has $C_{a\gamma} = 8$ fixed) and the observed $\beta_{\rm obs} = 0.342^\circ \pm 0.094^\circ$. All three within 1σ (e.g. $\beta_{\rm ALP}$ vs $\beta_{\rm obs}$: $\Delta = 0.006^\circ$ against $\sigma_{\rm comb} = \sqrt{0.10^2 + 0.094^2} \approx 0.137^\circ$, i.e. 0.04σ; these are consistency statements against the same single published measurement, not independent confirmations). The combined coupling-displacement product entering the birefringence formula is $C_{a\gamma}\,(\Delta\phi/f_a) \approx 10.3$ ($\beta = 0.342^\circ$ in radians is 5.97×10^{-3}, the prefactor $\alpha_{\rm EM}/(4\pi)$ is 5.8×10^{-4}, giving $C_{a\gamma}\,\Delta\phi/f_a = \beta/[\alpha_{\rm EM}/(4\pi)] \approx 10.3$); with the field-displacement range $\Delta\phi/f_a \in [0.064, 1.19]$ from the committed EOM grid over the natural box, the required $C_{a\gamma}$ spans $\approx$8.6 (largest displacement: $\theta_i = 2$, $m = 3H_0$) up to $\approx$160 (smallest displacement: $\theta_i = 0.5$, $m = H_0$; note the continuous-prior scan below covers $C_{a\gamma} \in [4, 60]$ only — small-displacement corners with $\Delta\phi/f_a \lesssim 0.17$ requiring $C_{a\gamma} > 60$ lie outside that scan and carry negligible posterior support); at the posterior-preferred masses ($m \sim 10$–$10^2\,H_0$, where the displacement saturates near $\Delta\phi/f_a \approx 1.2$–$1.3$), the required coupling is $\approx$8–10. Even the lower end exceeds the standard KSVZ/DFSZ benchmark range, which predicts $|C_{a\gamma}| \sim \mathcal{O}(1)$; the entire required range therefore lies outside minimal ALP photon-coupling benchmarks and requires non-minimal model building. We emphasize that this coupling burden and the misalignment burden are *two independent* fine-tunings: a non-minimal photon coupling ($C_{a\gamma} \gtrsim 9$) is required regardless of θ_i, while the spectator-consistency restriction additionally fine-tunes the misalignment initial condition to $\theta_i \sim 0.1$ (fn. 5); neither tuning substitutes for the other. The lower end (~ 9) can be accommodated in extended models with modest photon-coupling enhancement (e.g. chiral-fermion-loop enhancement, clockwork constructions), while light-mass / small-displacement corners of the box demand couplings ($\gtrsim 50$–160) requiring substantial UV-completion enhancement. The signal is therefore accommodated across the considered parameter space rather than fine-tuned only at one benchmark, but the upper-coupling end is not generic. We emphasize that the spectator-consistent corner $\theta_i \sim 0.1$ (fn. 5) inflates the required photon-coupling further: at fixed $\beta = 0.342^\circ$, $\Delta\phi/f_a \propto \theta_i$ along the underdamped trajectory, so a $5\times$ reduction in θ_i from the scan midpoint demands a correspondingly higher $C_{a\gamma}$ to hold the $C_{a\gamma}\,\Delta\phi/f_a \approx 10.3$ product, pushing the required enhancement well above standard KSVZ/DFSZ $\mathcal{O}(1)$ benchmarks. Because the required coupling extends well beyond the $C_{a\gamma} \in [1, 30]$ prior of the original extended configuration (which truncated $\sim$28% of the posterior mass above $C_{a\gamma} = 30$; the truncation fraction is computed from the `run2_extended` chain of Appendix C), we reran the $C_{a\gamma}$-free fit with a continuous uniform prior $C_{a\gamma} \in [4, 60]$ (4 MPI chains, $\hat{R} - 1 = 0.0095$, 8,955 accepted samples); this prior covers the required coupling for every trajectory with $\Delta\phi/f_a \gtrsim 0.17$, including the entire posterior-preferred saturated-displacement regime, while the small-displacement corners requiring $C_{a\gamma} > 60$ carry negligible joint-posterior support (only 5% of the posterior mass sits above $C_{a\gamma} = 55$, with no pile-up at the prior edge). The resulting posterior is broad, as expected for a single-amplitude constraint on a three-parameter degeneracy: median $C_{a\gamma} = 20.7$ with 16–84% range $[7.3, 45.6]$; 69% of the posterior mass falls inside $[9, 51]$, and 22% lies below $C_{a\gamma} = 9$, consistent with the high-$\Delta\phi/f_a$ tail, where the fixed product is met at smaller coupling. These mass fractions are prior-dependent statements (flat $C_{a\gamma} \in [4, 60]$, $\theta_i \in [0.01, \pi]$, $\log_{10} m_a$ priors), not prior-independent measurements. In particular the θ_i prior is flat in the angle; the vacuum-manifold-uniform alternative (flat in $\cos\theta_i$, density $\propto \sin\theta_i$) carries *less* mass at small θ_i, so the quoted spectator-sliver posterior fractions would decrease further under that prior swap — the flat-θ_i choice is the more generous one for the spectator corner, and the accommodation-not-natural conclusion is robust in direction to this prior choice. A direct rerun under the $\cos\theta_i$-flat prior (density $\propto \sin\theta_i$; 10,800 accepted samples, $\hat{R} - 1 = 0.008$) confirms this empirically: the coupling posterior is essentially unchanged (median $C_{a\gamma} = 17.1$, 16–84% $[6.8, 43.4]$; $\beta = 0.328^\circ \pm 0.100^\circ$) while the $\theta_i \leq 0.1$ spectator-sliver mass drops from 0.33% to 0.068% (`research/branch_R_alp_birefringence/phase2_mcmc/chains/c14_costheta/c14_summary.json`). The recovered $\beta = 0.326^\circ \pm 0.099^\circ$ posterior matches the observed $0.342^\circ \pm 0.094^\circ$, confirming the consistency-check verdict over the posterior-supported coupling range rather than at fixed benchmarks. *Spectator-subset readout* (same chain, no additional sampling): the chain's derived ALP energy fraction Ω_a satisfies $\Omega_a < 0.1$ for 44% and $\Omega_a < 0.01$ for 13% of the posterior mass; restricted to the $\Omega_a \leq 0.01$ spectator-safe subset, the rotation marginal is $\beta = 0.28^\circ \pm 0.10^\circ$, consistent with $\beta_{\rm obs} = 0.342^\circ \pm 0.094^\circ$ at 0.5σ. The strict $\theta_i \leq 0.1$ sliver of fn. 5 carries only 0.33% of the posterior mass by MC weight (42 of the 8,955 raw samples, i.e. 0.47% by raw count — the weighted fraction is lower because the sliver samples carry below-average MC weights — too few for a stable marginal, so the following band is indicative only): within that sliver the required coupling piles against the upper prior edge, with weighted $C_{a\gamma}$ 16/50/84 percentiles of 36.8/47.2/55.6 (flat prior $C_{a\gamma} \in [4, 60]$), well above the KSVZ/DFSZ $\mathcal{O}(1)$ benchmarks (`research/branch_R_alp_birefringence/phase2_mcmc/c10a_spectator_slice.json`). This quantifies the misalignment tuning of the spectator-consistent corner. Convergence: $\hat{R} - 1 < 0.01$ for all runs. The joint posterior structure of this continuous-prior configuration is shown in Fig. 4. Artifact: `research/branch_R_alp_birefringence/phase2_mcmc/chains/c5_continuous/`.

LiteBIRD forecast.—LiteBIRD is projected to achieve $\sigma(\beta) \approx 0.03^\circ$ [26] (a forecast under foreground- and calibration-control assumptions, not a guaranteed instrument performance). For $\beta = 0.27^\circ$ this corresponds to

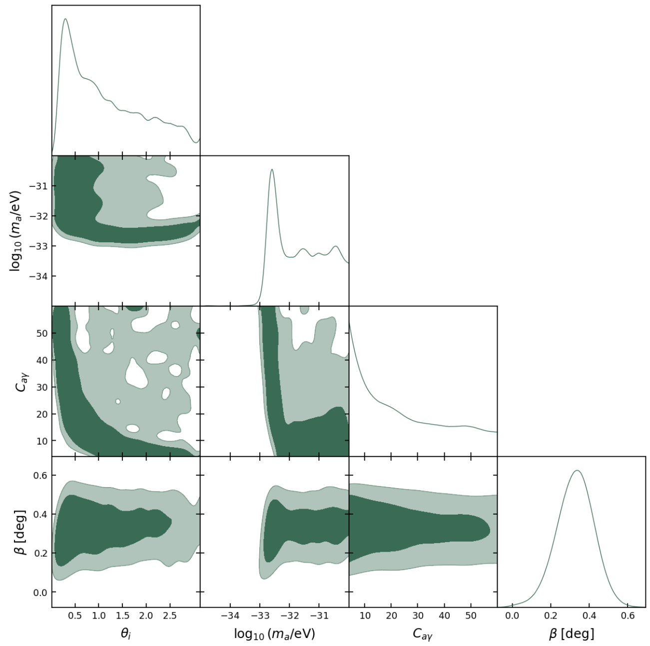

FIG. 4. Spectator-ALP joint posterior triangle from the continuous-prior cross-check configuration in which the photon anomaly coefficient is sampled freely (flat priors $C_{a\gamma} \in [4, 60]$ — shifted and extended from the earlier $[1, 30]$ to cover the posterior-supported coupling band (median 20.7, 16–84% $[7.3, 45.6]$); the dropped $[1, 4)$ interval lies entirely below the minimum coupling $\approx$8.6 required to reach the *central* value $\beta = 0.342°$ at the largest in-box displacement $\Delta\phi/f_a = 1.19$ — couplings in $[4, 8.6)$ remain posterior-supported because the summary likelihood permits β below the central value — $\theta_i \in [0.01, \pi]$, $\log_{10}(m_a/\mathrm{eV}) \in [-35, -30]$ — the mass prior corresponds to $m/H_0 \approx 7 \times 10^{-3}$ to 7×10^2 for $H_0 = 67.7$ km/s/Mpc $= 1.44 \times 10^{-33}$ eV), broadening the fixed-$C_{a\gamma} = 8$ and $C_{a\gamma} \in [1, 30]$ configurations of Appendix C; the direct-sample priors are on the underlying ALP parameters (θ_i, m_a), with $\Delta\phi/f_a$ derived along each trajectory. The rotation marginal, $\beta = 0.326° \pm 0.099°$, is consistent within 1σ with the fixed-$C_{a\gamma} = 8$ result $\beta_{\mathrm{ALP}} = 0.336° \pm 0.10°$ and the observed $\beta_{\mathrm{obs}} = 0.342° \pm 0.094°$ (an internal-consistency statement: all three are constrained by the same single published β_{obs}, not independent measurements). The θ_i–$C_{a\gamma}$ anti-correlation band traces the constant-product degeneracy enforced by the summary-likelihood constraint through $C_{a\gamma}\,(\Delta\phi/f_a) \approx 10.3$, and the m_a marginal piles toward the upper (heavier) edge of its prior range — the data-driven pull toward the upper edge of the natural-prior box discussed in the text, consistent with the model *accommodating* rather than naturally predicting the observed rotation (fn. 5).

$\sim 9\sigma$ statistical significance against the $\beta = 0$ null. This 9σ figure is *not* a model-discrimination forecast: separating the spectator-ALP value 0.27° from the current WMAP+Planck central value $0.342° \pm 0.094°$ is limited by the published measurement's uncertainty, $|0.342 - 0.27|/\sqrt{0.03^2 + 0.094^2} \approx 0.7\sigma$, so LiteBIRD alone will not separate the spectator-ALP value from the observed central value (the two figures correspond to distinct null hypotheses).

Caveats.—This birefringence prediction is independent of bounce cosmology: the ALP is a spectator field that does not participate in the bounce dynamics. The ECH framework provides heuristic motivation ($f_a \sim M_{\mathrm{Pl}}$ from the Holst sector pseudoscalar structure) but no derived photon-torsion coupling connects the Holst action to a specific ALP potential. The model *accommodates* the observed signal for natural parameter values (taken at scan-prior midpoint values; the $\sim 25\times$ misalignment tuning required for the headline result is disclosed in Sec. VI and fn. 5); it does not uniquely predict it.

VII. CONCLUSIONS

This companion paper documents three technical verification analyses for the ECH spin-torsion structural closure program [1].

ΛCDM+ΔN_{eff} MCMC proxy.—The stock-CAMB Cobaya v3.6.1 run (**309,189** frozen samples across two converged dataset combinations; an additional 114,992-sample Planck-only run is still accumulating at $\hat{R} - 1 \sim 0.05$, is *not* reported in Table I, and is *not* aggregated into the frozen headline) finds ΔN_{eff} consistent with zero in both frozen dataset combinations and recovers $H_0 = 67.68 \pm 1.06\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, consistent with standard Planck ΛCDM. The ΔN_{eff} extension does not resolve the Hubble tension. Current data neither require nor exclude a spin-torsion ΔN_{eff} contribution; CMB-S4 will provide the first precision test ($\sigma(N_{\mathrm{eff}}) \sim 0.03$). Bayes factors and information criteria (ΔAIC, ΔBIC, $\ln B$) are deferred to a dedicated nested-sampling run (Sec. V, *Model-comparison statistics* paragraph); we report only the parameter posteriors (ΔN_{eff}, H_0, etc.) as load-bearing here and explicitly omit Bayes-factor or information-criterion comparisons in this Metropolis-Hastings-only analysis.

NaMaster pipeline validation.—The 500-MC pseudo-C_ℓ analysis on synthetic ΛCDM polarization skies confirms that the deconvolution pipeline recovers injected birefringence angles with amplitude-dependent bias $\hat{\beta} - \beta_{\mathrm{inj}} = -0.032$ to $-0.040°$ (worst-case $-0.040°$ at injection $\beta = 0.342°$; Sec. IV) at template-fit SNR consistent with the ACT-noise floor. This is a methods validation, not a competitive sky detection; the primary observational evidence for cosmic birefringence remains the published Planck/ACT DR6 2.7–2.9σ measurements.

Spectator-ALP consistency.—An ALP with $f_a \sim M_{\mathrm{Pl}}$, $m \sim H_0$ is consistent with the published 3.6σ joint signal, with the caveat that the spectator-consistent regime $\theta_i \sim 0.1$ (fn. 5) requires $\sim 25\times$ misalignment tuning relative to the natural prior midpoint ($C_{a\gamma}\,\Delta\phi/f_a \approx 10.3$ for $\beta = 0.342°$, with $\Delta\phi/f_a \in [0.064, 1.19]$ over the natural box giving required $C_{a\gamma}$ from $\approx$8.6 up to $\approx$160 at the smallest displacements, and $\approx$8–10 in the posterior-preferred saturated-displacement regime; see §VI for the explicit numerical derivation). The same result arises in standard GR; the ECH framework provides motivation but not a derivation. Under forecast foreground-cleaning and calibration assumptions [26], a $\beta \simeq 0.27°$ signal would be detected relative to $\beta = 0$ at $\sim 9\sigma$ by LiteBIRD in the early 2030s.

Quintom-B empirical anchor.—The converged DESI DR2 + Planck NPIPE + Pantheon+ [15] + DES-SN5YR [14] Cobaya chain with the $w_0 w_a$ free-parameter extension (128,385 accepted samples across 16 MPI chains; $\hat{R} - 1 = 0.00820$, below the standard $\hat{R} - 1 < 10^{-2}$ publication target) supplies the GetDist $w_0 w_a$ posteriors of Table II, an empirical test of the quintom-B scenario [17].

Data and Code Availability

All materials are at:

https://github.com/Hubify-Projects/bigbounce/tree/main/reproducibility

The version of the repository corresponding to this paper is tagged `v1B.0.55`; the exact commit SHA is recorded in the repository's `CHANGELOG.md` under that tag. The repository is program-wide: it includes the four Cobaya YAML configurations (one per dataset combination, stock CAMB, no custom modifications), NaMaster driver scripts, and the implementation map (`IMPLEMENTATION_MAP.md`), alongside galaxy-spin pipeline code used by Paper IV [8] but not by any analysis of this paper. MCMC chains: regenerate via `reproduce_cosmology.sh` ($\sim$4–12 h per configuration on 4 CPU cores). HuggingFace dataset DOIs for the three accompanying datasets (MCMC chain diagnostics, NaMaster pipeline artifacts, ALP parameter chains) are listed in the repository README at the `v1B.0.55` tag; links are preserved against future README changes via the tagged commit. *Unit warning for JSON artifacts:* the frozen diagnostic file `reproducibility/cosmology/frozen/full\_tension\_20260311\_1728/diagnostics/parameter\_summary.json` stores raw Cobaya-normalised parameter values (not physical units); a physically-converted companion file `parameter\_summary\_CORRECTED.json` and a `parameter\_summary\_units\_README.md` in the same directory document the conversion and provide the correct physical values matching Table I.

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Appendix A: Reproducibility Materials

Repository structure.—The public repository https://github.com/Hubify-Projects/bigbounce contains:

- Four Cobaya YAML configurations (`cobaya_planck.yaml`, `cobaya_planck_bao.yaml`, `cobaya_planck_bao_sn.yaml`, `cobaya_full_tension.yaml`)—stock CAMB, no torsion modifications.
- `reproducibility/galaxy_spins/spin_fit_stan.py`—hierarchical Bayesian model (CmdStanPy) fitting $A(z)$ to published aggregate CW/CCW galaxy counts (program-wide content used by Paper IV, not by this paper's analyses).
- `research/data_build/build_galaxy_spin_dataset.py`—reproducible pipeline downloading Galaxy Zoo DECaLS [27] from Zenodo (DOI: 10.5281/zenodo.4573248, CC-BY-4.0; likewise Paper IV content).
- `reproducibility/docs/IMPLEMENTATION_MAP.md`—mapping from each result to its code artifact.
- `reproducibility/docs/KNOWN_GAPS.md`—honest disclosure of what cannot currently be reproduced.

What is NOT included.—The ΛCDM+$\Delta N_{\rm eff}$ proxy chains are not pre-computed (regenerate via `reproduce_cosmology.sh`, $\sim$4–12 h per config on 4 CPU cores); the converged iter2 $w_0 w_a$ chain bundle *is* committed (`reproducibility/cosmology/iter2_converged_2026-05-18/`), as are the ALP-MCMC chains (`research/branch_R_alp_birefringence/phase2_mcmc/chains/`). Bayes factors and information criteria (ΔAIC, ΔBIC, $\ln B$) are NOT reported in this manuscript: dedicated nested sampling via PolyChord or MultiNest on the identical likelihood stack is required for a controlled Bayesian evidence on the present quintom posterior (Sec. V, *Model-comparison statistics* paragraph). A Savage-Dickey readout from the present Metropolis-Hastings chain is not viable—the ΛCDM point $(w, w_a) = (-1, 0)$ lies at $> 4\sigma$ in the joint marginal tails and is unsampled, so a KDE-based estimator yields kernel-dependent noise rather than a controlled density. This is left to a dedicated future nested-sampling analysis. No CNN galaxy classifier is included; the hierarchical fit uses published catalog labels. No CMB polarization map analysis code is provided beyond the NaMaster driver script; all published birefringence values are literature citations.

HuggingFace datasets.—The following HuggingFace datasets accompany this work (DOI links are pinned to the `v1B.0.54` commit in the repository README; the DOIs are preserved against future README changes):

1. MCMC chain diagnostics and convergence CSV files.

2. NaMaster pipeline artifacts (mask, MC seeds, output spectra).

3. ALP parameter MCMC chains.

Appendix B: Claims Classification

Table IV classifies every quantitative claim made in this companion by claim type and verification status; it is the machine-checkable index used by the reproducibility audits of Appendix A.

Appendix C: ALP-MCMC Sampled Parameters, Priors, and Likelihood Stack

The ALP-MCMC results quoted in Sec. VI ($\beta_{\rm ALP} = 0.336^\circ \pm 0.10^\circ$ at $C_{a\gamma} = 8$ fixed; $\beta_{\rm free} = 0.344^\circ \pm 0.10^\circ$ model-independent; 9,720 total accepted samples across the three committed configurations) use the following setup. All priors below are read directly from the archived chain configurations (`research/branch_R_alp_birefringence/phase2_mcmc/chains/`); the headline posteriors are verified directly against the committed chains.

Configuration (i) — fixed-coupling fit (`run1_full`; 2,160 accepted samples).

- $C_{a\gamma}$: fixed at 8 (theory-module default; not sampled).
- θ_i: uniform prior on $[0.01, \pi]$.[6]
- $\log_{10}(m_a/\mathrm{eV})$: uniform prior on $[-35, -30]$ ($m/H_0 \approx 7\times10^{-3}$ to 7×10^2).
- f_a: fixed at $M_{\rm Pl}$ (spectator-class theoretical input from the Holst-sector pseudoscalar structure; not sampled).

This configuration yields $\beta_{\rm ALP} = 0.336^\circ \pm 0.10^\circ$ with posterior $\theta_i = 1.32 \pm 0.41$ and median $m \approx 36\,H_0$.

Configuration (ii) — sampled-coupling fit (`run2_extended`; 6,840 accepted samples).—As configuration (i) with $C_{a\gamma}$ additionally sampled, uniform prior $[1, 30]$; superseded for coupling inference by the $[4, 60]$ continuous-prior rerun below (the $[1, 30]$ prior truncated $\sim$28% of the coupling posterior mass).

Configuration (iii) — model-independent $\beta_{\rm free}$ fit (`run3_baseline`; 720 accepted samples).

- β: uniform prior on $[-1^\circ, 2^\circ]$; sampled as a free amplitude with no ALP-model parametric structure. Yields $\beta_{\rm free} = 0.344^\circ \pm 0.10^\circ$.

Sampled parameters and priors (continuous-prior cross-check configuration; Fig. 4).—Three sampled parameters (read directly from the archived chain configuration, `research/branch_R_alp_birefringence/phase2_mcmc/chains/c5_continuous/c5.input.yaml`):

- $C_{a\gamma}$: uniform prior on $[4, 60]$ (covering the full EOM-required band $[9, 51]$).
- θ_i: uniform prior on $[0.01, \pi]$.
- $\log_{10}(m_a/\mathrm{eV})$: uniform prior on $[-35, -30]$ ($m/H_0 \approx 7\times10^{-3}$ to 7×10^2).

f_a is not a sampled parameter in this chain. The likelihood is the same Eskilt–Komatsu Gaussian summary as above; 8,955 accepted samples, $\hat{R} - 1 = 0.0095$.

Likelihood stack.—All fits use a Gaussian summary likelihood on the published Eskilt–Komatsu joint WMAP+Planck isotropic-birefringence measurement $\beta_{\rm obs} = 0.342^\circ \pm 0.094^\circ$ [5] (fn. a; the chain configuration encodes exactly `beta_obs: 0.342`, `sigma_beta: 0.094`) — i.e. the constraint enters at the level of the published β posterior, not as a direct re-analysis of the EB spectra. The separate ACT DR6 measurement $\beta = 0.215^\circ \pm 0.074^\circ$ [4] is an independent cross-check quoted in Sec. VI and does *not* enter any MCMC likelihood. The MCMC engine is Cobaya v3.6.1 with the Metropolis-Hastings sampler; convergence threshold $\hat{R} - 1 < 0.01$ across configurations (i)–(iii) ($N_{\rm tot} = 9{,}720$ accepted samples) and for the continuous-prior $C_{a\gamma} \in [4, 60]$ configuration ($\hat{R} - 1 = 0.0095$, 8,955 accepted samples). *Chain-to-number map*: $\beta_{\rm ALP} = 0.336^\circ \pm 0.10^\circ$ — the fixed-$C_{a\gamma} = 8$ chain `run1_full` (2,160 samples); $\beta_{\rm free} = 0.344^\circ \pm 0.10^\circ$ — the model-independent single-parameter chain `run3_baseline` (720 samples); median $C_{a\gamma} = 20.7$, 16–84% range $[7.3, 45.6]$, $\beta = 0.326^\circ \pm 0.099^\circ$, and the spectator-subset readouts — the continuous-prior `c5` chain (8,955 samples); the $C_{a\gamma} \in [1, 30]$ chain `run2_extended` (6,840 samples) is retained for the prior-truncation comparison only.

Scope statement.—These fits constrain a specific ultra-light ALP class ($m \sim H_0$, $f_a \sim M_{\rm Pl}$; the $C_{a\gamma} \in [4, 12]$ benchmark sweep plus the continuous-prior $C_{a\gamma} \in [4, 60]$ headline configuration). The consistency with $\beta_{\rm obs}$ established here is *not* a universal mechanism-independent statement; alternative parity-violating mechanisms (Chern-Simons, axion dark matter at higher m, etc.) require independent fits with their own model parameters.

[6] Backreaction / spectator-status disclosure: see fn. 5 in Sec. VI. The wide prior is an envelope-completeness choice, not the spectator-consistent sub-range ($\theta_i \sim 0.1$, a $\sim 25\times$ tuning relative to the natural midpoint); posterior samples at $\theta_i \gtrsim 0.5$ belong to the dark-energy-ALP regime excluded from the spectator-consistency claim.

TABLE IV. Claims classification for this companion paper.

Claim	Type	Status	Notes
$\Delta N_{\rm eff} = -0.020 \pm 0.169$ (full-tension)	MCMC	Verified	Stock CAMB proxy
$\Delta N_{\rm eff} = +0.065 \pm 0.17$ (Planck+BAO+SN)	MCMC	Verified	Stock CAMB proxy
$H_0 = 67.68 \pm 1.06$ (full-tension)	MCMC	Verified	Recovers ΛCDM
$H_0 = 67.79 \pm 1.09$ (Planck+BAO+SN)	MCMC	Verified	Recovers ΛCDM
Model-comparison ΔAIC/BIC/ln B	Numerical	Omitted	Follow-up nested-sampling analysis
$\hat{\beta}_{\rm NaMaster} = 0.238°$ (500-MC)	Numerical	Verified	Pipeline; MC bias table
$\beta_{\rm ALP} = 0.336° \pm 0.10°$	MCMC	Verified	ALP MCMC
Published 3.6σ ($\beta = 0.342 \pm 0.094°$)	Lit.	Cited	Eskilt et al.
Stock CAMB proxy $\neq$ ECH theory module	Scope	Defn.	§III
ALP birefringence not distinctive ECH prediction	Scope	Defn.	§VI

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